

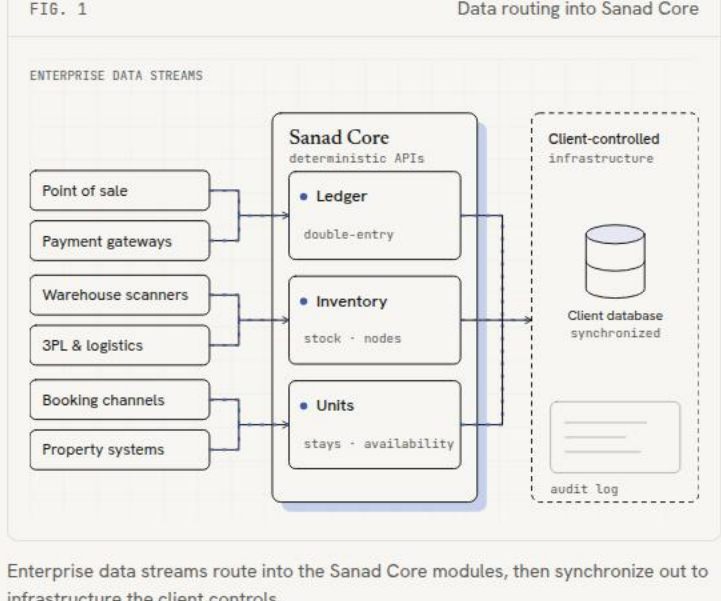
Enterprise engineering studio · Intelligent software for modern commerce

Intelligent systems built to quietly run commerce.

Sanad constructs operational backbones and bespoke autonomous logic for commercial enterprises. We remove the cognitive and administrative weight of scaling — engineering a structural reduction of at least 30% in daily operational load.

Request Technical Dossier

Read the 30% baseline



Enterprise data streams route into the Sanad Core modules, then synchronize out to infrastructure the client controls.

For Leadership

Predictable velocity, margin protection, and operational clarity.

For Engineering

Decoupled architecture, local-first integrity, and zero black-box dependencies.

01 Proof by metric

The 30% efficiency baseline

Operational load is not a feeling. It is waiting, re-keying and reconciling — three things that can be measured, and removed. Every system we build is specified against one baseline: a structural reduction of at least 30% in daily operational load.

Workstream	Before	With Sanad
01 Inventory & Supply Sync	18 hours of multi-channel lag between stock and sales	Sub-second synchronization across every channel
02 Financial & Reconciled Auditing	Repeated reconciliations, performed by staff each cycle	Zero-loss balancing routines, automated end to end
03 Human Error Overhead	Re-keyed data entered again in each system it touches	Eliminated structurally: one entry, one record of truth

Definition

$L = t_{wait} + t_{rekey} + t_{reconcile}$
daily operational load, in hours

$L' \leq 0.70 \times L$
the 30% Law

Daily operational load, indexed

Baseline 100

With Sanad ≤ 70 At least 30% reclaimed

02 Systems & architecture

Ready-made modules. Bespoke engineering.

Two explicit offerings, never blended. Pillar A is production-tested software, configured for your business. Pillar B is engineered from scratch, for operational flows that belong to you alone.

Pillar A

A

Turnkey Intelligent Modules

Pre-configured SaaS

High-efficiency, production-tested systems, ready for deployment.

A.1 Inventory & Distributed Warehousing Logic

One stock position across every warehouse, store and channel.

Interface wireframe

Schema

```
type StockLevel {
  sku      string
  node     NodeId // warehouse or store
  on_hand  int
  reserved int // held for open orders
  version  lamport // deterministic ordering
}
```

Topology: n-node, multi-warehouse

Sync: event-sourced, deterministic merge

Reservations: per-channel holds, auto-expiry

Offline: node-local ledger, reconcile on reconnect

Latency: < 100 ms in-site, sub-second across channels

A.2 Hospitality & Unit Management Engines

Units, stays and turnover, held to a single source of availability.

Interface wireframe

Schema

```
type Stay {
  unit      UnitId
  from      date
  to        date
  status    vacant | occupied | turnover | blocked
  rate_plan RatePlanId
}
```

// invariant: no two stays overlap on one unit

Allocation: overlap-free by construction

Housekeeping: status transitions create tasks

Channels: booking channels kept in step

Rates: plans, seasons and restrictions

Handover: turnover opens at check-out

A.3 Unified Commerce & POS Ledger Sync

Every sale, online or in store, posts once to one ledger.

Interface wireframe

Schema

```
type LedgerEntry {
  tx      TxId // idempotency key
  account AccountCode
  debit   decimal
  credit  decimal
  source  pos | web | erp | gateway
}
```

// invariant: Σ debit = Σ credit, per transaction

Model: double-entry, append-only

Idempotency: one key per transaction

Gateways: settlement matched automatically

Close: daily close with variance report

Audit: immutable history, exportable

Pillar B

B

Tailored Commercial Architecture

Bespoke systems

From-scratch systems, engineered for unique, proprietary operational flows.

B.1 Custom internal operating systems

Replacing patchwork spreadsheets

One system of record in place of a folder of spreadsheets: role-based views, enforced workflows, and a history of every change.

record single source of truth

access role-based

history append-only

B.2 Private agentic workflows

Operating securely on client infrastructure

Autonomous logic that runs inside your perimeter, on your hardware or your cloud account. Every action is logged, bounded by explicit permissions, and reversible.

runtime client infrastructure

egress denied by default

oversight approval gates, full action log

B.3 High-availability distributed databases

With offline-first synchronization

Replicated stores that keep accepting writes when the network does not, then converge deterministically when it returns.

writes local-first

merge deterministic

single point of failure none, by design

03 System mechanics & technical rigor

Specified like infrastructure, stated so it can be tested.

For CTOs and technical evaluators

Data ownership
Sovereign
Complete sovereignty. Databases synchronize with client-controlled infrastructure.
Reliability
Offline-capable
Operational nodes designed to function without internet disruption.
Latency
< 100 ms
Sub-100ms internal state changes across multi-terminal setups.

Representative event trace Illustrative sequence, not a benchmark.

```
# multi-terminal sale on the shop floor
t+808 ms term-02 sale.created tx=A-10441
t+808 ms node-01 ledger.posted xdr = xcr ok
t+819 ms node-01 stock.reserved sku=1042 -1
t+837 ms term-01 state.applied v=2291
t+841 ms term-03 state.applied v=2291

# network lost: the node keeps working locally
t+912 ms term-02 sale.created tx=A-10442
t+921 ms node-01 ledger.posted queued=1

# network restored: replicate to client-db
t+4.2 s node-01 replicate converged v=2292
```

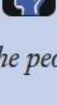
Target state changes under 100 ms across terminals, with no dependency on the network.

Integrations
Deterministic
Clean, deterministic APIs connecting legacy ERPs, payment gateways, and third-party logistics.

The Sanad manifesto

Software should clarify the mind, not demand it.

Most enterprise software creates its own bureaucracy. Teams spend hours managing the tool that was supposed to save them time. Sanad is built on the inverse thesis: intelligent software works silently in the background so leaders and builders can dedicate their focus to what actually matters.



Support for the people who lead.

04